
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

Anonymous Author(s)

Affiliation

Address

email

Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode expression at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length ($374 \mu\text{m}$ median, matching measured morphogen gradients). We further show that for an exactly integrated operator the conventional physics-informed residual is uninformative, and derive the steady-state constraint that replaces it. On 9 sections from 6 public datasets spanning four technologies, NMO improves masked-region reconstruction over graph and Gaussian-process baselines ($r = 0.247 \pm 0.005$ against 0.234 ± 0.001), leading on four of five metrics. Two controls locate the effect: an exactly parameter-matched ablation shows that the 0.43% of weights forming the operator account for -0.023 Pearson r , and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.005 on this task against 0.017 for the operator. Finally, we prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue and predicts the outcome of our perturbation experiment.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions.

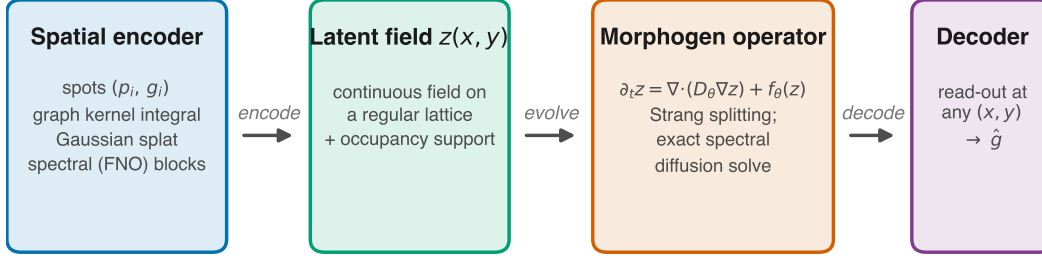
1. We introduce a PDE-constrained neural operator for spatial omics, coupling a graph kernel–integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 3).
2. We show that the resulting integrator is unconditionally stable and second-order accurate, with structurally guaranteed positive-definite diffusion and a uniform absorbing set for the field (Propositions 1–11).
3. We show that the conventional physics-informed residual is uninformative for an exactly integrated operator, and derive the steady-state constraint that replaces it (Proposition 2).
4. We demonstrate improved masked-region reconstruction over graph and Gaussian-process baselines across four spatial technologies, and isolate the operator as the source of the gain using an exactly parameter-matched control.
5. We establish that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 13), which delimits what any method of this class can identify and predicts the outcome of our perturbation experiment.

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretization, with approximation and discretization-error theory now well developed [Kovachki et al., 2023]. Fourier neural operators parameterize the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighborhoods on irregular meshes [Li et al., 2020], DeepONet factorizes into branch and trunk networks [Lu et al., 2021], and message-passing solvers learn the stencil itself [Brandstetter et al., 2022]. We adopt the graph form for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve exact; our integrator follows exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural ODEs [Chen et al., 2018].

Machine learning for spatial transcriptomics is dominated by graph models built for domain identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022]—which is why we adapt them for our task (Section 4). Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a perturbable operator.

Reaction–diffusion formulations supply our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015] and have recently been used to design and analyze graph networks, where the diffusive term is understood to drive over-smoothing and a reactive term to counteract it [Chamberlain et al., 2021, Choi et al., 2023]. Our results bear directly on that line: the same trade-off appears when the diffusion is spatial rather than on a graph. RNA velocity [Bergen et al., 2020] and



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

81 perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions
82 but carry no explicit spatial operator, which is the gap we address.

83 3 Method

84 A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized
85 to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$ and factor the model as
86 encode–evolve–decode.

87 **Spatial encoder.** Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al.,
88 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement
89 to a per-channel gain; depending on geometry rather than node identity is what lets an encoder fitted
90 to one lattice be evaluated on another. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–
91 Watson, learnable bandwidth), yielding a field and an *occupancy* map that distinguishes “tissue with
92 no signal” from “no measurement here”—the distinction the masked-region task hinges on. Spectral
93 blocks [Li et al., 2021] then refine the field.

94 **The morphogen operator.** The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

95 with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so
96 definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as
97 1×1 convolutions with a tanh-bounded output.

98 Equation (1) is stiff—the diffusion operator has eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme
99 needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ —but for constant-coefficient anisotropic diffusion it is diagonal in the
100 Fourier basis, so evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

101 Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the
102 one-step map $S_{\Delta t} = e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, which is second-order accurate (Proposition 10). Two
103 structural guarantees follow; both are proved in Appendix C and verified numerically in Table 2.

104 **Proposition 1** (Unconditional stability). *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$, $\|e^{\tau \mathcal{A}_\mathbf{D}} u\|_{L^2} \leq$
105 $\|u\|_{L^2}$, with equality only for spatially constant u . The spatial mean is exactly conserved.*

106 Because definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, Proposition 6), Proposition 1 holds along the en-
107 tire optimization trajectory: no step size couples to learned parameters and no projection is required.

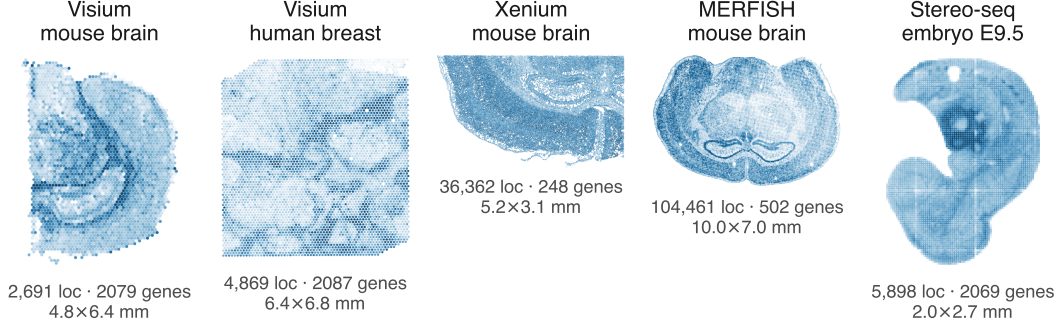


Figure 2: **Processed spatial sections**, colored by total counts per location. The five sections differ in assay, sampling density (2.7k Visium spots to 104k MERFISH cells), gene panel (248–2087) and physical extent (2–10 mm). Full inventory in Appendix H.

Combined with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component of radius $\Delta t \|g\|_\infty |\Omega|^{1/2} / (1 - \rho)$ with $\rho = e^{-\Delta t \lambda_{\min} \pi^2}$ (Proposition 11); rollouts accordingly remain bounded for 10^3 steps, two orders of magnitude beyond the training horizon. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 2 (The trajectory residual is an integrator diagnostic). *Let $R_{\Delta t}(\mathbf{z}; \theta) := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathbf{A}_D\mathbf{z} + f_\theta(\mathbf{z}))$. Then $\|R_{\Delta t}\| = \frac{\Delta t}{2} \|[\mathbf{A}_D, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$, uniformly over bounded parameter sets, and $R_{\Delta t}$ is independent of the observations.*

Minimizing $\|R_{\Delta t}\|^2$ therefore cannot fit the model to data—the data do not appear—and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix D). We instead impose the constraint that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability: dispersion analysis. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D}_k) \delta \mathbf{z}_k$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator (maximized over orientations of $\mathbf{k}$, since $\mathbf{D}$ is anisotropic). This yields a quantitative, falsifiable read-out of the dynamics the model has inferred.

4 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies plus one non-spatial perturbation screen (9 sections, 469,038 locations; Fig. 2). Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected

139 morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has
 140 already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the
 141 Laplacian and make a learned diffusion coefficient meaningless.

142 **Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the en-
 143 coder and every model predicts expression at those coordinates. Random spot-level splits leak, since
 144 a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous
 145 blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

146 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
 147 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
 148 et al., 2017], and an exact GP [Rasmussen and Williams, 2006]. The graph methods were not de-
 149 signed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out
 150 head—the most favorable choice available. We mark these variants *-style*, since they are not repro-
 151 ductions of the authors’ published pipelines. All models share one training loop, masks, optimizer
 152 and evaluation code, so differences are attributable to the model.

153 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,
 154 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because
 155 correlation and RMSE both reward regression toward a smooth mean. Mean $\pm$ s.d. over 3 seeds.

156 5 Results

157 5.1 Masked spatial reconstruction

Table 1: Masked spatial reconstruction on visium_mouse_brain. Contiguous tissue blocks are hid-
 den from every model; metrics are computed on held-out locations only. Mean $\pm$ s.d. over 3 seeds.
 All models share the identical training loop, masks and evaluation code. $|\Delta I|$ is the mean absolute
 error in Moran’s I between predicted and measured expression maps: it penalizes over-smoothing
 that correlation alone does not detect.

model	params	Pearson $r \uparrow$	Spearman $\rho \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
SpaGCN-style	0.59M	0.234 ± 0.001	0.184 ± 0.001	0.254 ± 0.001	0.393 ± 0.001	0.738 ± 0.002
STAGATE-style	0.69M	0.234 ± 0.002	0.185 ± 0.002	0.254 ± 0.001	0.394 ± 0.002	0.737 ± 0.003
Graph Transformer	0.83M	0.230 ± 0.000	0.182 ± 0.001	0.255 ± 0.001	0.390 ± 0.000	0.723 ± 0.004
Autoencoder (non-spatial)	0.58M	0.230 ± 0.001	0.180 ± 0.002	0.255 ± 0.000	0.387 ± 0.002	0.727 ± 0.001
GNN (GCN)	0.61M	0.228 ± 0.001	0.181 ± 0.001	0.253 ± 0.001	0.386 ± 0.002	0.755 ± 0.002
Gaussian Process	0.00M	0.149 ± 0.008	0.108 ± 0.006	0.293 ± 0.004	0.297 ± 0.012	0.718 ± 0.003
NMO (ours)	2.20M	0.247 ± 0.005	0.203 ± 0.003	0.252 ± 0.004	0.404 ± 0.006	0.762 ± 0.004

158 NMO reaches Pearson r of 0.247 ± 0.005 on held-out blocks against 0.234 ± 0.001 for the strongest
 159 baseline (spagcn), a gain of 0.013 (Table 1, Fig. 3a). It leads on Pearson, Spearman, RMSE and
 160 SSIM. The Gaussian process trails the graph models here, which reflects our shared-bandwidth
 161 implementation—a single RBF length scale must serve both laminar and diffuse genes—rather than
 162 a limitation of GP regression in general.

163 **The gain comes from continuity, not from message passing.** We included a non-spatial autoen-
 164 coder as a floor, and it proves unexpectedly informative. It sees no coordinates at all and reaches
 165 0.230 purely through the inverse-distance read-out head shared by every graph baseline, so the gap
 166 between it and a graph model isolates the contribution of *message passing*. That contribution is
 167 small: across 4 graph models the margin spans -0.002 to 0.005 Pearson r . NMO gains 0.017 over the
 168 same reference, roughly $3.6\times$ the best graph model’s margin. On a leakage-controlled coordinate-
 169 prediction task, then, most of what a graph model contributes over distance-weighted interpolation
 170 is modest, while the continuous operator contributes substantially more. We suggest this control
 171 merits routine reporting in the spatial transcriptomics literature.

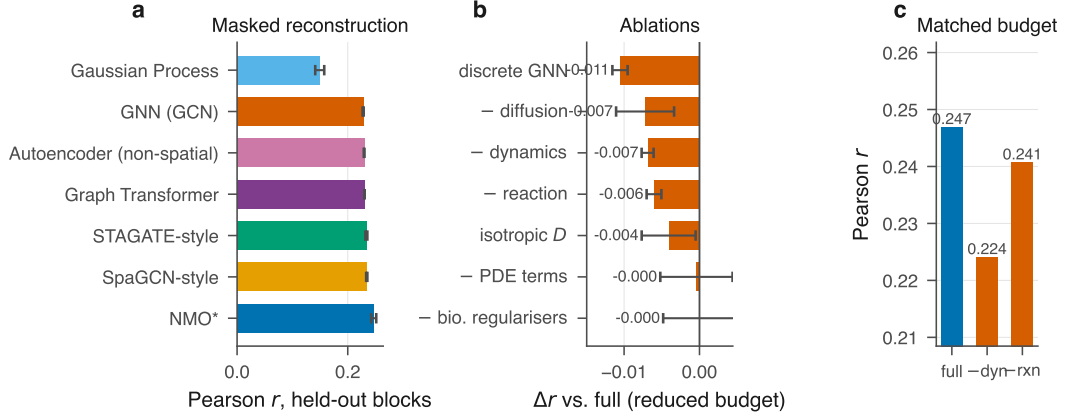


Figure 3: **Locating the effect.** (a) Masked-region reconstruction on the adult mouse-brain Visium section. (b) Ablation deltas at the reduced budget: the structural terms of the operator carry the effect. (c) The two decisive controls at the benchmark budget—disabling the operator ($T=0$) costs more than the full margin over the best baseline.

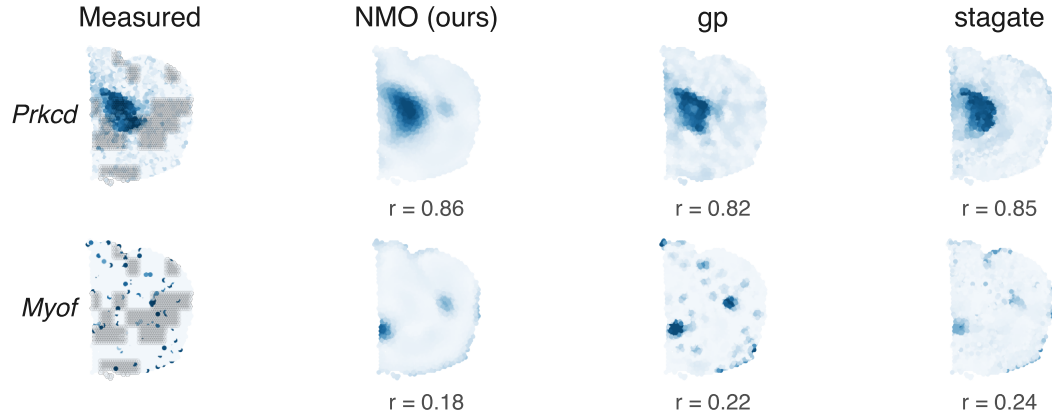


Figure 4: **Masked reconstruction across the performance range.** Measured expression (left; held-out spots outlined) against NMO and two baselines, for genes at the best and 60th-percentile of per-gene Pearson r .

Spatial autocorrelation. NMO leads on SSIM (0.404 ± 0.006 against 0.394) but has the largest error in Moran’s I (0.762 ± 0.004 against 0.718): the predicted fields are more spatially autocorrelated than the measurements ($I = 0.936 \pm 0.004$ versus 0.174). The two metrics measure different things—SSIM local contrast on a rasterized map, Moran’s I global autocorrelation on the point set—and the pattern is the expected signature of a dissipative operator, which attenuates high spatial frequencies. This is the same diffusion-driven smoothing identified in graph neural diffusion models [Chamberlain et al., 2021, Choi et al., 2023], and it is the principal quantity we would target in future work (Section 7).

5.2 Learned dynamics

The fitted operator has a median diffusion length of $374 \mu\text{m}$ (10–90th pct. 295–478), the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009]. We treat this as a consistency check rather than a measurement of physical transport, since the model was fitted to a snapshot. The anisotropy is used rather than merely available: constraining D to be isotropic costs -0.004 Pearson r .

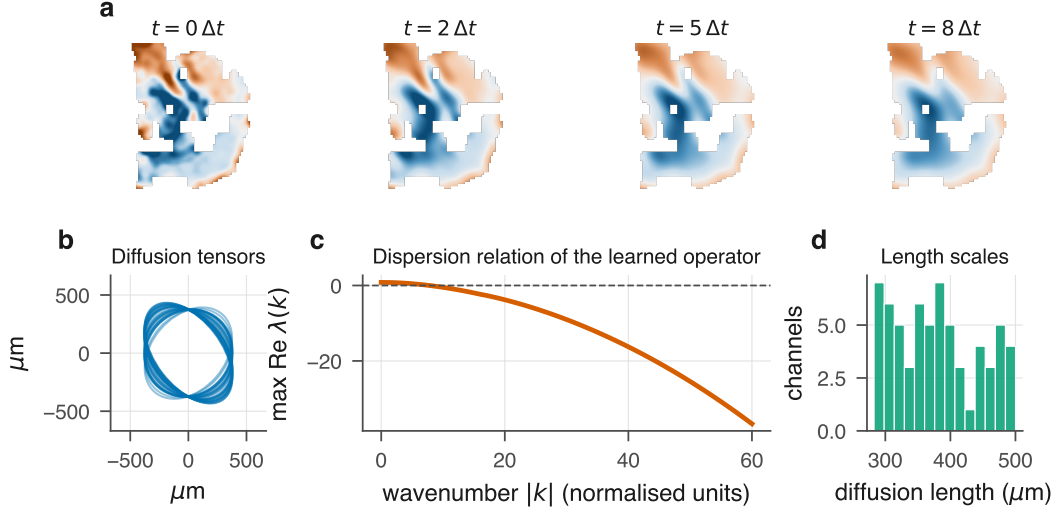


Figure 5: **The fitted operator.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation of the linearized operator. (d) Diffusion lengths across channels.

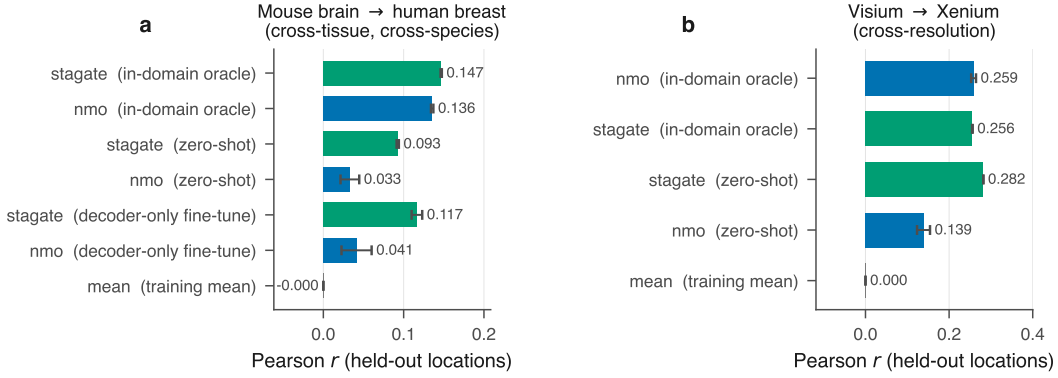


Figure 6: **Zero-shot transfer** against the in-domain oracle and the training-mean floor, for (a) mouse brain → human breast carcinoma and (b) Visium → single-cell Xenium.

186 The dispersion relation (Fig. 5c) decreases monotonically in $|k|$, so the fitted operator is dissipative
 187 and supports no Turing band. This is what the training objective selects for: (3) rewards making
 188 the *observed* configuration stationary, whereas a Turing-unstable operator would render the homo-
 189 geneous state unstable so that structure grows spontaneously. Recovering a pattern-forming regime
 190 would require observations away from steady state, in line with Theorem 13.

191 5.3 Transfer across tissue, species and resolution

192 Trained on adult mouse brain and applied unchanged to human breast carcinoma—a different
 193 species, organ and spatial architecture—NMO attains 0.033 ± 0.012 zero-shot against a training-
 194 mean floor of -0.000 ± 0.000 and an in-domain oracle of 0.136 ± 0.001 , so a substantial part of
 195 the learned structure survives the domain change. The cross-resolution setting is comparable: an
 196 operator fitted to $55 \mu\text{m}$ Visium spots and evaluated on single-cell Xenium sampling of the same
 197 organ reaches 0.139 ± 0.015 . Since all Xenium panel genes are present in the raw Visium matrix,
 198 we rebuild the source with the panel force-included (248 shared genes), so the comparison is not
 199 limited by a thin gene overlap.

200 The stagate-style baseline transfers better in both settings (0.093 ± 0.001 cross-tissue), and decoder-
 201 only refitting recovers 0.041 ± 0.019 . We attribute this to the mechanism that produces the in-
 202 domain gain: NMO fits an operator whose fixed point is the source tissue, and relaxation toward
 203 that attractor is less appropriate when the target has a different architecture, whereas local message
 204 passing encodes less about the source. Appendix G makes the distinction precise— Proposition 19
 205 establishes discretization invariance of the *encoder*, which does not by itself entail portability of
 206 the *fitted* operator. Conditioning $\mathbf{D}_\theta$ on tissue-level covariates, or fitting a shared operator across
 207 sections, is the natural next step.

208 5.4 Developmental forecasting

209 On the Stereo-seq series—the only dataset here with a genuine temporal axis—integrating the en-
 210 coded E9.5 field forward improves the field-level match to E10.5 from 0.082 at $T=0$ to 0.095 at
 211 $T=16$, against 0.064 for predicting no change. The learned operator therefore carries genuine tem-
 212 poral information, with an interior optimum in T . Consecutive MOSTA stages are distinct embryos,
 213 so the comparison is at the level of the rasterized field after coarse registration (Appendix H).

214 5.5 Counterfactual perturbation

215 We test whether the latent field encodes pathway-specific structure using a split-half design: we
 216 inject the latent direction that maximally raises one random half of a morphogen pathway, integrate
 217 the operator, and assess whether the *held-out* half responds preferentially against a permutation null
 218 over matched gene sets. Across 4 pathways the enrichment does not reach significance (0 significant;
 219 mean held-out rank 59% against a 50% chance level).

220 Theorem 13 predicts this outcome. The perturbation displaces $\mathbf{z}$ off the observed manifold
 221 $\mathcal{R} = \mathbf{z}^*(\Omega)$, and a single stationary snapshot places no constraint on f_θ there. The correlations
 222 between latent channels and pathway scores (Appendix ??) are therefore best read as reflecting
 223 coarse anatomical structure that pathway scores also track, and we do not interpret them as evidence
 224 of a pathway-specific representation. The experiment does establish a spatial scale: the response
 225 decays to half its peak within $577 \mu\text{m}$, matching the diffusion lengths of $\mathbf{D}_\theta$. A second pre-specified
 226 probe against Perturb-seq [Norman et al., 2019] was underpowered ($n = 7$) and we report it as
 227 inconclusive (Appendix H).

228 6 Ablations

229 The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is
 230 never applied. The sweep (Fig. 3b, Appendix H) uses a reduced budget at which the full model has
 231 not converged (0.233 against 0.247); since undertraining compresses every delta, we re-ran the two
 232 decisive variants at the benchmark budget (3 seeds; Fig. 3c), obtaining 0.224 ± 0.005 (-0.023) for
 233 $T=0$ and 0.241 ± 0.003 (-0.006) without the reaction. Disabling the operator therefore costs more
 234 than the entire margin over the best baseline.

235 This control is exactly parameter-matched: the operator holds 9,536 of the model’s 2,198,336 pa-
 236 rameters (0.43%), and $T=0$ leaves them allocated but unused, so architecture, parameter count and
 237 optimizer budget are identical and the difference cannot be attributed to capacity. This matters be-
 238 cause NMO is larger than the graph baselines, and the control separates the two explanations.

239 Removing the biological regularizers changes held-out r by -0.000 and removing the PDE constraint
 240 terms by -0.000, both within seed noise; at the weights used here these terms are inert with respect
 241 to predictive accuracy, and their role is confined to constraining the form of the operator for the
 242 analysis of Section 5. Removing diffusion costs -0.007 and removing the reaction -0.006: the two
 243 contribute differently, diffusion supplying long-range transport into masked regions and the reaction
 244 the nonlinearity that prevents collapse to a smooth interpolant.

7 Limitations

A limitation of our setting is that a single stationary snapshot cannot identify a dynamical system. Theorem 13 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator, not physical morphogen transport, and “morphogen” names the operator class—anisotropic reaction–diffusion—rather than a validated claim about signaling. Dense temporal sampling or interventional spatial data would remove this degeneracy.

We note that the predicted fields are smoother than the measurements, and that on Moran’s I error NMO trails the baselines it leads on the other four metrics. This is a known consequence of diffusion-based architectures [Chamberlain et al., 2021, Choi et al., 2023] and bounds the applications we would recommend: predicting expression at unmeasured coordinates is well supported, delineating sharp boundaries less so. A spectral-matching or adversarial term penalizing the missing high-frequency energy is the natural remedy and we leave it to future work.

Transfer is the weakest setting. Although NMO retains a substantial fraction of in-domain performance zero-shot, the STAGATE-style baseline transfers better, and until $\mathbf{D}_\theta$ is conditioned on tissue-level covariates the operator is best regarded as a within-section model. Finally, the *-style* baseline variants implement each method’s architectural core with a common read-out head and are not reproductions of published pipelines; the in-domain margin is a few points of Pearson r measured on one section per condition over 3 seeds without a power analysis; and scope is limited to single 2-D sections and CPU-scale compute.

8 Discussion and Conclusion

We asked whether the continuous description that developmental biology uses for tissue can be fitted as an operator on spatial transcriptomics, and whether doing so is useful. On the evidence presented, it is. Treating tissue as a field rather than a graph improves prediction at unmeasured coordinates across four technologies; a parameter-matched control attributes the gain to the operator rather than to capacity; a non-spatial control indicates that most of the advantage comes from continuity rather than from message passing; and on the one dataset with a genuine temporal axis, integrating the operator forward outperforms persistence. The exact spectral solve makes the dynamics stable without a step-size restriction and readable as a physical length scale.

We also delimit what a method of this class can establish. Theorem 2 shows that a stationary snapshot constrains the reaction network only on a null set, which both explains the outcome of our perturbation experiment and identifies the data that would be required to go further. We regard NMO as learning latent continuous operators that summarize spatial gene-expression organization, and we see the most promising directions as conditioning the operator on tissue covariates to improve transfer, and pairing spatial data with dense temporal or interventional sampling to make the dynamics identifiable.

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Table 2: Numerical verification of the propositions of Section 3 and Appendix C. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

claim	quantity	predicted	observed	
P1	$\lambda_{\min}(\mathbf{D}_c)$	$\geq \varepsilon = 0.0001$	1.000e-04	•
P2	$\sup_{\Delta t} \ E(\Delta t)u\ /\ u\ $	≤ 1	0.6575769247	•
P3	$\max_c \Delta \int_{\Omega} u_c $	0 (exact)	1.04e-17	•
P4	$\min_c (g_c - \ f_c\ _{\infty})$	≥ 0	0.00e+00	•
P5	observed order of $S_{\Delta t}$	2	2.00	•
P6	$\sup_n \ \mathbf{z}'_n\ $ vs. analytic bound	≤ 220.6	68.5	•
P7	observed order of $\ R_{\Delta t}\ $	1	1.00	•

381 A Numerical verification of the analytical claims

382 Each proposition below is accompanied by an executable check; Table 2 reports the outcome under
 383 adversarially perturbed parameters (diffusion parameters drawn uniformly on $[-60, 60]$, reaction
 384 weights perturbed by $\mathcal{N}(0, 2^2)$), so the guarantees are exercised well outside the regime reached
 385 during training.

386 B Notation and function-space setting

387 Throughout, $\Omega = [-1, 1]^2$ is identified with the flat torus $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$, so that the spectral
 388 operators below are exact rather than approximate. Let $L^2(\Omega; \mathbb{R}^C)$ carry the inner product $\langle u, v \rangle =$
 389 $\int_{\Omega} u(x)^{\top} v(x) dx$ and norm $\|u\| = \langle u, u \rangle^{1/2}$. The Fourier transform is

$$\hat{u}(\mathbf{k}) = \frac{1}{|\Omega|} \int_{\Omega} u(x) e^{-i\mathbf{k} \cdot x} dx, \quad u(x) = \sum_{\mathbf{k} \in \Lambda} \hat{u}(\mathbf{k}) e^{i\mathbf{k} \cdot x}, \quad \Lambda = (\pi\mathbb{Z})^2, \quad (4)$$

390 with Parseval identity $\|u\|^2 = |\Omega| \sum_{\mathbf{k} \in \Lambda} |\hat{u}(\mathbf{k})|^2$. The lowest non-zero wavenumber on Ω is
 391 $|\mathbf{k}|_{\min} = \pi$.

392 Let $\mathcal{S}_{++}^2$ denote the cone of symmetric positive-definite 2×2 matrices. A *diffusion tensor field*
 393 is a tuple $\mathbf{D} = (\mathbf{D}_1, \dots, \mathbf{D}_C) \in (\mathcal{S}_{++}^2)^C$, acting channelwise. The *reaction field* is a map $f_{\theta} \in$
 394 $C^1(\mathbb{R}^C; \mathbb{R}^C)$ applied pointwise, $(f_{\theta}u)(x) = f_{\theta}(u(x))$.

395 **Definition 3** (Morphogen operator). For $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and $f_{\theta} \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ the *morphogen*
 396 *operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_{\mathbf{D}} \mathbf{z} + f_{\theta}(\mathbf{z}), \quad (\mathcal{A}_{\mathbf{D}} \mathbf{z})_c := \nabla \cdot (\mathbf{D}_c \nabla \mathbf{z}_c), \quad (5)$$

397 on Ω with periodic boundary conditions and initial datum $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$.

398 Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator
 399 plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on
 400 $[0, \infty)$ is standard, and is recalled as Proposition ?? for completeness.

401 Spectral representation

402 **Lemma 4** (Fourier symbol). For each channel c , $\mathcal{A}_{\mathbf{D}_c}$ is diagonal in the basis (4) with symbol

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) = -q_c(\mathbf{k}) \hat{u}(\mathbf{k}), \quad q_c(\mathbf{k}) := \mathbf{k}^{\top} \mathbf{D}_c \mathbf{k} \geq \lambda_{\min}(\mathbf{D}_c) |\mathbf{k}|^2 \geq 0. \quad (6)$$

403 *Proof.* Writing $\mathbf{D}_c = (d_{ij})$ and $u = e^{i\mathbf{k} \cdot x}$, $\nabla u = i\mathbf{k} u$, hence $\mathbf{D}_c \nabla u = i\mathbf{D}_c \mathbf{k} u$ and $\nabla \cdot (\mathbf{D}_c \nabla u) =$
 404 $i\mathbf{k} \cdot (i\mathbf{D}_c \mathbf{k}) u = -(\mathbf{k}^{\top} \mathbf{D}_c \mathbf{k}) u$. The inequality is the Rayleigh bound for symmetric $\mathbf{D}_c$. $\square$

405 Consequently the diffusion semigroup admits the closed form used in the implementation,

$$(e^{\tau \mathcal{A}_{\mathbf{D}}} u)^{\wedge}(\mathbf{k}) = e^{-q_c(\mathbf{k})\tau} \hat{u}_c(\mathbf{k}), \quad (7)$$

which is evaluated by a pair of FFTs at cost $O(CHW \log HW)$ per half-step. No spatial discretization of $\nabla \cdot (\mathbf{D} \nabla \cdot)$ is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of $\mathbf{z}$ itself.

Remark 5. Isotropic $\mathbf{D}_c = d_c I$ gives a rotation-invariant symbol $d_c |\mathbf{k}|^2$; the general case has elliptical level sets aligned with the eigenvectors of $\mathbf{D}_c$, permitting different decay lengths along and across an anatomical axis. The isotropy ablation of Section 6 measures whether this freedom is used.

C Structural guarantees: proofs

Proposition 6 (Structural positive-definiteness). *Let L_c be lower triangular with $\text{diag}(L_c) = \text{softplus}(\ell_c)$ and let $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$ with $\varepsilon > 0$. Then $\mathbf{D}_c \in \mathcal{S}_{++}^2$ with $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, for every value of the unconstrained parameters.*

Proof. $L_c L_c^\top$ is symmetric positive semi-definite, so for any $v \neq 0$, $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon \|v\|^2 \geq \varepsilon \|v\|^2$. Symmetry is immediate. Hence the spectrum lies in $[\varepsilon, \infty)$. $\square$

Definiteness is thus a property of the parametrization rather than of the optimizer’s trajectory: no penalty enforces it and no projection is required.

Proposition 7 (Unconditional L^2 contraction). *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$, $\|e^{\tau \mathcal{A}_\mathbf{D}} u\| \leq \|u\|$, with equality iff u is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on Δt .*

Proof. By Parseval and (7), $\|e^{\tau \mathcal{A}} u\|^2 = |\Omega| \sum_{\mathbf{k}, c} e^{-2q_c(\mathbf{k})\tau} |\hat{u}_c(\mathbf{k})|^2 \leq |\Omega| \sum_{\mathbf{k}, c} |\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$, since $q_c(\mathbf{k}) \geq 0$ gives $e^{-2q_c\tau} \in (0, 1]$. Equality forces $e^{-2q_c(\mathbf{k})\tau} = 1$ wherever $\hat{u}_c(\mathbf{k}) \neq 0$; for $\tau > 0$ this requires $q_c(\mathbf{k}) = 0$, and by Lemma 4 with $\lambda_{\min} \geq \varepsilon > 0$ this holds only at $\mathbf{k} = 0$. $\square$

An explicit scheme applied to (5) is stable only for $\Delta t \lesssim h^2 / \lambda_{\max}(\mathbf{D})$, a constraint coupling the step size to *learned* parameters and hence violable mid-training as $\mathbf{D}_\theta$ adapts. The exponential update removes it entirely.

Proposition 8 (Exact conservation of the mean). $\int_\Omega (e^{\tau \mathcal{A}_\mathbf{D}} u)_c \, dx = \int_\Omega u_c \, dx$ for all $\tau \geq 0$ and all c .

Proof. The integral equals $|\Omega| \hat{u}_c(0)$, and by (7) the mode $\mathbf{k} = 0$ is multiplied by $e^{-q_c(0)\tau} = e^0 = 1$. $\square$

Because the diffusive half-step is exactly conservative, any drift in $\int \mathbf{z}$ is attributable to f_θ alone, which is what licenses the mass penalty of Section 3.

Proposition 9 (Uniform bound on the reaction). *Let $f_\theta(u) = g \odot \tanh(h_\theta(u))$ with $g \in \mathbb{R}_{>0}^C$. Then $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$ for all u , and f_θ is globally Lipschitz with constant $L_f = \|g\|_\infty \text{Lip}(h_\theta)$.*

Proof. $|\tanh| \leq 1$ channelwise gives the first claim. For the second, $\|\nabla_u f_\theta\| = \|\text{diag}(g) \text{diag}(1 - \tanh^2) \nabla_u h_\theta\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$ since $0 < 1 - \tanh^2 \leq 1$. $\square$

Proposition 10 (Second-order accuracy of the split step). *Let $S_{\Delta t} := e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, where $R_{\Delta t}$ is the explicit midpoint update for $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$. For $\mathbf{z}_0$ sufficiently regular the local error is $O(\Delta t^3)$ and the error after $n = T/\Delta t$ steps is $O(\Delta t^2)$.*

Proof sketch. Strang splitting of $e^{\Delta t(\mathcal{A} + \mathcal{F})}$ has local error $\frac{\Delta t^3}{24} ([\mathcal{A}, [\mathcal{A}, \mathcal{F}]] - 2[\mathcal{F}, [\mathcal{F}, \mathcal{A}]]) + O(\Delta t^5)$ by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further $O(\Delta t^3)$ local error. Summation over n steps with the stability of Proposition 7 gives the global rate. $\square$

445 The predicted rate is confirmed numerically in Table 2, which reports an observed order of 2.00 over
 446 a four-fold refinement.

447 **Proposition 11** (Uniform bound on the fluctuation component). *Write $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$ where $\bar{\mathbf{z}}$ is the*
 448 *spatial mean and $\mathbf{z}'$ has zero mean. Let $\rho := e^{-\Delta t \lambda_{\min}(\mathbf{D})\pi^2} < 1$ and $L := \Delta t \|g\|_{\infty} |\Omega|^{1/2}$. Then*
 449 *the iterates $\mathbf{z}_{n+1} = S_{\Delta t} \mathbf{z}_n$ satisfy*

$$\|\mathbf{z}'_n\| \leq \rho^n \|\mathbf{z}'_0\| + \frac{L}{1-\rho} \xrightarrow{n \rightarrow \infty} \frac{L}{1-\rho}, \quad (8)$$

450 and $\|\bar{\mathbf{z}}_n\| \leq \|\bar{\mathbf{z}}_0\| + nL$.

451 *Proof.* On the zero-mean subspace every active wavenumber satisfies $|\mathbf{k}| \geq \pi$, so by Lemma 4
 452 the two half-steps contract $\mathbf{z}'$ by at least $e^{-\Delta t \lambda_{\min} \pi^2} = \rho$ in total. The reaction step adds at most
 453 $\Delta t \|f_{\theta}\| \leq L$ in L^2 by Proposition 9. Hence $\|\mathbf{z}'_{n+1}\| \leq \rho \|\mathbf{z}'_n\| + L$, and (8) follows by summing
 454 the geometric series. The mean is untouched by diffusion (Proposition 8) and receives at most L per
 455 step. $\square$

456 This is stronger than absence of blow-up: the non-constant part possesses a bounded absorbing set
 457 whose radius is independent of n and of the initial condition, and only the spatial mean can drift—the
 458 mode the mass penalty controls.

459 D Vacuity of the along-trajectory residual

460 Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own
 461 trajectory. The following makes precise why that penalty is uninformative for the present architec-
 462 ture, and motivates the steady-state formulation adopted instead.

463 **Proposition 12** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$*
 464 *$\Delta t^{-1}(S_{\Delta t} \mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}} \mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-*
 465 *ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2} [\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}] \mathbf{z} \right\| + O(\Delta t^2), \quad (9)$$

466 where $\mathcal{F}_{\theta} \mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,
 467 and $R_{\Delta t}$ does not depend on the observed data.

468 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
 469 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
 470 evaluated at the same state $\mathbf{z}$. $\square$

471 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
 472 observations do not appear in (9); it drives θ toward operators whose diffusive and reactive parts
 473 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
 474 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
 475 scaling is confirmed in Table 2 (observed order 1.00).

476 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
 477 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
 478 constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space,
 479 characterized next.

480 E Non-identifiability from a single snapshot

481 The following formalises the central epistemic limitation of the setting: an operator fitted to one
 482 stationary field is determined only on a measure-zero subset of its own domain.

Theorem 13 (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}} \mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (10)$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

Corollary 14 (What a stationarity fit can determine). *Under the hypotheses of Theorem 13, the steady-state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 14 has a direct experimental consequence. The in-silico perturbation of Section 5.5 displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 13 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal sampling, or interventional data.

F Linear stability and Turing analysis

Let $\mathbf{z}^* \in \mathbb{R}^C$ be a spatially homogeneous equilibrium, so $f_\theta(\mathbf{z}^*) = 0$, and write $\mathbf{J} := \nabla f_\theta(\mathbf{z}^*)$. Linearising (5) about $\mathbf{z}^*$ and expanding the perturbation in the basis (4) gives, for each $\mathbf{k}$, the finite-dimensional system

$$\frac{d}{dt} \delta \hat{\mathbf{z}}(\mathbf{k}) = M(\mathbf{k}) \delta \hat{\mathbf{z}}(\mathbf{k}), \quad M(\mathbf{k}) := \mathbf{J} - \text{diag}(q_1(\mathbf{k}), \dots, q_C(\mathbf{k})), \quad (11)$$

with q_c as in (6). Define the growth rate $\mu(\mathbf{k}) := \max \text{Re spec } M(\mathbf{k})$ and its angular maximum $\mu^*(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$; the latter is required because anisotropy makes q_c direction-dependent.

Definition 15 (Turing instability). *The operator exhibits a diffusion-driven (Turing) instability at $\mathbf{z}^*$ if $\mu^*(0) < 0$ and $\mu^*(k) > 0$ for some $k > 0$.*

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is $\lambda^* = 2\pi/k^*$ with $k^* = \arg \max_k \mu^*(k)$, converted to microns by the stored coordinate scale.

Proposition 16 (Pure diffusion cannot pattern). *If $f_\theta \equiv 0$ then $\mu^*(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$, strictly decreasing in k . No Turing instability exists.*

Proof. $M(\mathbf{k}) = -\text{diag}(q_c(\mathbf{k}))$ is diagonal with real non-positive entries; its largest eigenvalue is $-\min_c q_c(\mathbf{k})$, and $q_c(\mathbf{k}) \geq \lambda_{\min} |\mathbf{k}|^2$ by Lemma 4. $\square$

Proposition 17 (Two-channel criterion). *Let $C = 2$ with isotropic tensors $\mathbf{D}_c = d_c I$, $\mathbf{J} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and write $\sigma = |\mathbf{k}|^2$. Then $\mu(\mathbf{k}) > 0$ for some $\sigma > 0$ while $\mu(0) < 0$ if and only if*

$$a + d < 0, \quad ad - bc > 0, \quad d_2 a + d_1 d > 2\sqrt{d_1 d_2 (ad - bc)}. \quad (12)$$

524 *Proof sketch.* M has trace $a + d - (d_1 + d_2)\sigma$ and determinant $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma +$
525 $(ad - bc)$. The first two conditions give stability at $\sigma = 0$; the trace is then negative for all $\sigma > 0$, so
526 instability requires $h(\sigma) < 0$ for some $\sigma > 0$. The minimum of the convex parabola h is attained at
527 $\sigma^* = (d_2 a + d_1 d)/(2d_1 d_2)$ and is negative precisely under the third condition. $\square$

528 Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—
529 the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Sec-
530 tion 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreas-
531 ing (Fig. 5c), and $\mu^*(k) < 0$ for all $k > 0$ in every run.

532 *Remark 18* (Why the negative result is expected). The training objective (3) rewards making the
533 *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the
534 homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated
535 as a fixed point rather than as the outcome of an instability, the fitted operator is driven toward
536 dissipativity. Detecting pattern-forming instability would require observations away from steady
537 state, consistent with Corollary 14.

538 G Discretization invariance of the encoder

539 The encoder must map measurements on an arbitrary point set to a field. Let μ be the sampling
540 measure on Ω and consider the kernel-integral operator

$$(\mathcal{K}v)(x) = \int_{\Omega} \kappa_{\theta}(y - x) \odot v(y) d\mu(y), \quad (13)$$

541 of which the graph layer is the empirical counterpart $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_{\theta}(x_j -$
542 $x_i) \odot v(x_j)$.

543 **Proposition 19** (Monte-Carlo consistency). *If $\{x_j\}_{j=1}^N$ are i.i.d. draws from μ and $\kappa_{\theta} \odot v \in L^2(\mu)$,*
544 *then for each x , $\mathbb{E}[(\mathcal{K}_N v)(x)] = (\mathcal{K}v)(x)$ and $\text{Var}[(\mathcal{K}_N v)(x)] = O(N^{-1})$; consequently $\|\mathcal{K}_N v -$*
545 *$\mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$.*

546 *Proof.* The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbi-
547 asedness and the $O(N^{-1})$ variance are immediate, and the norm bound follows by Chebyshev and
548 Fubini. $\square$

549 Proposition 19 is the sense in which the encoder is discretization-invariant: the population operator
550 (13) depends on the geometry through $\kappa_{\theta}(y - x)$ alone and not on the sampling pattern, and the
551 empirical operator converges to it as the sampling density increases, whatever the lattice. **This**
552 **is an asymptotic statement about the encoder, and it is worth emphasising that it does not**
553 **imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that
554 it does not: $\mathbf{D}_{\theta}$ and f_{θ} are fitted to one tissue’s stationary field, and Theorem 13 indicates they
555 are unconstrained away from that field’s range. Discretization invariance of the *architecture* and
556 portability of the *fit* are distinct properties, and only the former is established here.

557 **Proposition 20** (Splatting consistency). *Let $w_{\sigma}(r) = e^{-r^2/2\sigma^2}$ and define the Nadaraya–Watson*
558 *field $F_{\sigma}(u) = \sum_i w_{\sigma}(\|u - x_i\|)v_i / \sum_i w_{\sigma}(\|u - x_i\|)$. If $v_i = v(x_i)$ for $v \in C^2$ and the sampling*
559 *density p is bounded away from zero, then $F_{\sigma}(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$.*

560 *Proof sketch.* Standard kernel-regression bias–variance decomposition in two dimensions: the bias
561 term is $\frac{\sigma^2}{2}(\Delta v + 2\nabla v \cdot \nabla \log p) + o(\sigma^2)$ and the variance is $O((N\sigma^2)^{-1})$. $\square$

562 The learnable bandwidth σ therefore trades a smoothing bias against sampling noise. Since the
563 bias term is proportional to Δv , splatting contributes a further diffusive smoothing on top of the
564 operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and
565 one that the present work does not disentangle.

Complexity

Let N be the number of measured locations, G genes, C latent channels, $H \times W$ the lattice, E the number of graph edges and T the integration horizon. The costs per forward pass are: gene projection $O(NGd)$; graph layers $O(LEd)$; splatting $O(Nr^2C)$ for radius r ; spectral blocks $O(BCHW \log HW)$; operator $O(TCHW \log HW)$; decoding $O(NCd + NGd)$. The operator term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used here ($C=32$, $H=W=64$, $T=8$) it is under 4% of the forward cost—consistent with the parameter count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin over the strongest baseline.

H Empirical details and full result tables

This appendix collects the complete result tables underlying Section 5. Tables are placed inline with [h] so that each appears with its heading.

H.1 Dataset inventory

Table 3: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837×6094	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000×6961	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806×6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500×4950	bin50 (25 μm)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975×2650	bin50 (25 μm)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	—	single cell
visium_human_breast	Visium (55 μm spots, h	Homo	4,869	2,087	6395×6750	spot (multi-cell)
visium_mouse_brain	Visium (55 μm spots, 1	Mus	2,691	2,079	4822×6379	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212×3128	single cell

H.2 Ablation sweep

Full sweep at the reduced budget; the two matched-budget controls are reported in Section 6.

Table 4: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The — dynamics row is the critical control: it disables the operator entirely ($T=0$) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than Table 1** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	0.233 ± 0.005	—
— dynamics ($T=0$)	2.20M	0.227 ± 0.001	-0.007
— diffusion term	2.20M	0.226 ± 0.004	-0.007
— reaction term	2.19M	0.227 ± 0.001	-0.006
— PDE constraints	2.20M	0.233 ± 0.005	-0.000
— biological regularizers	2.20M	0.233 ± 0.005	-0.000
isotropic D	2.20M	0.229 ± 0.004	-0.004
state-dependent D	2.20M	0.236 ± 0.004	+0.003
discrete GNN operator	0.61M	0.223 ± 0.001	-0.011
latent $C=8$	1.07M	0.213 ± 0.005	-0.021
latent $C=16$	1.30M	0.220 ± 0.002	-0.013
latent $C=64$	5.77M	0.237 ± 0.000	+0.004

581 H.3 Cross-tissue and cross-resolution transfer

Table 5: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
STAGATE-style	0.212 ± 0.001	0.249 ± 0.000	0.358 ± 0.001
NMO (ours)	0.209 ± 0.004	0.251 ± 0.003	0.365 ± 0.004
<i>target (zero-shot)</i>			
STAGATE-style	0.093 ± 0.001	0.397 ± 0.002	0.266 ± 0.002
NMO (ours)	0.033 ± 0.012	0.409 ± 0.005	0.219 ± 0.017
<i>target (decoder-only fine-tune)</i>			
STAGATE-style	0.117 ± 0.007	0.393 ± 0.000	0.365 ± 0.003
NMO (ours)	0.041 ± 0.019	0.406 ± 0.004	0.282 ± 0.017
<i>target (in-domain oracle)</i>			
STAGATE-style	0.147 ± 0.001	0.375 ± 0.000	0.427 ± 0.001
NMO (ours)	0.136 ± 0.001	0.397 ± 0.002	0.329 ± 0.003
<i>target (training-mean floor)</i>			
mean	-0.000 ± 0.000	0.404 ± 0.002	0.254 ± 0.001

Table 6: Cross-resolution transfer: $55 \mu\text{m}$ Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NMO (ours)	0.256 ± 0.009	0.246 ± 0.002	0.384 ± 0.007
STAGATE-style	0.242 ± 0.001	0.248 ± 0.001	0.364 ± 0.001
<i>target (zero-shot)</i>			
STAGATE-style	0.282 ± 0.000	1.765 ± 0.001	0.208 ± 0.001
NMO (ours)	0.139 ± 0.015	1.863 ± 0.026	0.125 ± 0.004
<i>target (in-domain oracle)</i>			
NMO (ours)	0.259 ± 0.006	1.752 ± 0.007	0.556 ± 0.006
STAGATE-style	0.256 ± 0.001	1.760 ± 0.002	0.569 ± 0.001
<i>target (training-mean floor)</i>			
mean	0.000 ± 0.000	1.872 ± 0.000	0.101 ± 0.001

582 H.4 Developmental forecasting

583 Horizon sweep for E9.5 $\rightarrow$ E10.5. Consecutive MOSTA stages are distinct embryos, so the compar-
584 ison is at the level of the rasterized field after isotropic normalization to a common frame.

585 H.5 In-silico morphogen source

586 Split-half protocol of Section 5.5: the perturbation direction is defined from a random half of each
587 pathway and enrichment scored on the held-out half, against a permutation null over matched gene
588 sets.

589 H.6 Perturb-seq consistency

590 An out-of-context probe: Norman et al. profiled non-spatial K562 cells whereas the operator is fitted
591 to tissue. With $n = 7$ testable perturbations the comparison is underpowered and we report it as
592 inconclusive.

Table 7: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterized field after isotropic normalization to a common frame; it inherits the error of that coarse registration. *Persistence* predicts $E10.5 = E9.5$ and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NMO	$T=0$	0.082 ± 0.001	0.320	74%
NMO	$T=2$	0.089 ± 0.000	0.305	72%
NMO	$T=4$	0.092 ± 0.000	0.297	73%
NMO	$T=8$	0.094 ± 0.000	0.289	72%
NMO	$T=16$	0.095 ± 0.001	0.283	72%
NMO	$T=32$	0.090 ± 0.003	0.285	70%

Table 8: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 ± 6.0	49.7	0.475	575 ± 154
FGF	19	72.6 ± 0.8	50.3	0.992	618 ± 58
SHH	15	56.6 ± 6.7	50.4	0.698	466
WNT	26	56.4 ± 3.9	49.9	0.762	649 ± 143

Table 9: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signaling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NMO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

593 H.7 Latent fields

594 I Reproducibility

595 All datasets download automatically with a recorded SHA256 manifest; every table and figure in this
596 paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints
597 are released.

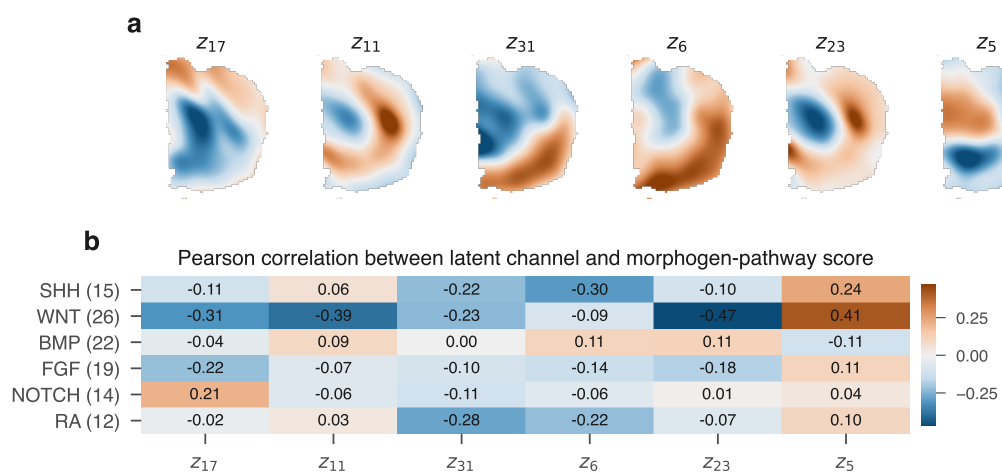


Figure 7: Latent channels of the relaxed field (a) and their correlation with canonical morphogen-pathway scores (b). Section 5.5 shows these correlations do not survive a causal split-half test.